Pizza Sales Dashboard — Power BI Project Report

1. Introduction

The **Pizza Sales Dashboard** is an interactive business intelligence dashboard developed using **Microsoft Power BI**. The purpose of this project is to analyze pizza sales data and provide meaningful insights into revenue, orders, quantity sold, product performance, and sales trends.

The dashboard converts raw sales data into interactive visualizations that help users understand overall business performance and identify the best and worst-performing pizzas.

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2. Project Objective

The main objectives of this project are:

- To analyze total pizza sales and revenue.
- To understand the number of orders and pizzas sold.
- To identify the best and worst-performing pizzas.
- To analyze sales performance by pizza category and size.
- To identify sales trends over time.
- To compare the performance of different pizza products.
- To create an interactive and user-friendly Power BI dashboard.

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3. Dataset Description

The dataset contains information related to pizza orders, pizza products, prices, categories, sizes, and order details.

The dataset consists of four main tables:

Orders

Contains information about individual orders, including:

- Order ID
- Order Date
- Order Time

Order Details

Contains details about the pizzas included in each order:

- Order ID
- Pizza ID
- Quantity

Pizzas

Contains pizza-level information:

- Pizza ID
- Pizza Type ID
- Size
- Price

Pizza Types

Contains descriptive information about pizzas:

- Pizza Type ID
- Pizza Name
- Category
- Ingredients

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4. Data Model

The tables were connected in Power BI using relationships between common IDs.

The main relationships are:

- **Orders → Order Details** using order_id
- **Pizzas → Order Details** using pizza_id
- **Pizza Types → Pizzas** using pizza_type_id

This relational model allows the dashboard to combine order, product, category, size, quantity, and revenue information.

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5. Dashboard Pages

The dashboard contains four main pages:

1. **Overview**
2. **Products**
3. **Time Analysis**
4. **Performance**

Interactive navigation buttons are provided on the sidebar to move between the four pages.

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6. Overview Page

The **Overview** page provides a high-level summary of the pizza business.

Key KPIs

- Total Revenue
- Total Orders
- Total Pizzas Sold
- Average Order Value
- Average Pizzas per Order

Visualizations

The page includes:

- **Monthly Revenue Trend**
- **Revenue by Pizza Category**
- **Revenue by Pizza Size**
- **Top 10 Pizzas by Revenue**

This page helps users quickly understand the overall sales performance of the business.

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7. Product Analysis Page

The **Products** page focuses on individual pizza performance.

Key Metrics

- Total Pizza Types
- Best Pizza
- Total Quantity Sold
- Best Category

Visualizations

The page contains:

- **Top 10 Pizzas by Revenue**
- **Top 10 Pizzas by Quantity**
- Product/category analysis

- Size-based quantity analysis
- Pizza performance details

This page helps identify which pizza products contribute most to sales and which products require further attention.

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8. Time Analysis Page

The **Time Analysis** page focuses on understanding sales patterns over time.

Key Metrics

- Peak Hour
- Peak Day
- Peak Month
- Total Orders

Visualizations

The page includes:

- **Orders by Hour**
- **Revenue by Day**
- **Orders by Day**
- **Monthly Sales Trend**

This analysis helps identify when customer demand is highest and can support better staffing, inventory planning, and promotional decisions.

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9. Performance Page

The **Performance** page provides a detailed comparison of the strongest and weakest performers.

Key Metrics

- Top Pizza
- Bottom Pizza
- Top Category
- Lowest Category

Visualizations

- **Top 10 Pizzas by Revenue**
- **Bottom 10 Pizzas by Revenue**

- **Detailed Pizza Performance Table**

The detailed table contains:

Category | Pizza | Size | Orders | Quantity | Revenue

This page helps management identify products that are performing well and products that may need improvement.

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10. Key DAX Measures

Several DAX measures were created to calculate important business metrics.

Total Orders

```
Total Orders = DISTINCTCOUNT(orders[order_id])
```

Total Quantity

```
Total Quantity =SUM(order_details[quantity])
```

Total Revenue

```
Total Revenue =SUMX(
   order_details,
   order_details[quantity] * RELATED(pizzas[price])
)
```

Average Order Value

```
Average Order Value =DIVIDE(
   [Total Revenue],
   [Total Orders]
)
```

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11. Business Insights

The dashboard can be used to identify several important business insights, such as:

- Which pizzas generate the highest revenue.
- Which pizzas have the highest sales volume.
- Which pizza category performs best.
- Which pizza sizes contribute most to revenue.
- Which days have higher customer demand.

- Which periods show stronger sales activity.
- Which products are underperforming.

These insights can help the business improve product planning, inventory management, staffing, and marketing strategies.

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12. Recommendations

Based on the dashboard analysis, the business can:

- Promote high-performing pizzas to increase revenue.
- Review low-performing pizzas and consider improving or replacing them.
- Maintain sufficient inventory for popular pizza sizes and categories.
- Increase staffing during peak sales periods.
- Use customer demand trends to plan promotions.
- Focus marketing efforts on products with strong sales potential.

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13. Conclusion

The **Pizza Sales Dashboard** demonstrates how Power BI can transform raw sales data into meaningful and interactive business insights.

By combining data modeling, DAX calculations, KPIs, charts, tables, and interactive navigation, the dashboard provides a complete view of pizza sales performance.

The four dashboard pages — **Overview, Products, Time Analysis, and Performance** — allow users to analyze the business from different perspectives and support data-driven decision-making.

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14. Tools & Technologies

Tools Used:

- Kaggle— Data Source
- Microsoft Power BI — Data Modeling & Visualization
- DAX — Calculated Measures and Analysis

Dashboard Features:

- Interactive charts
- KPI cards
- Top/Bottom analysis

- Product-level analysis
- Time-based analysis
- Page navigation
- Interactive filtering